Clinical Note Intelligence with RAG

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Healthcare providers spend hours reviewing unstructured notes - we turn that into seconds.

The Problem

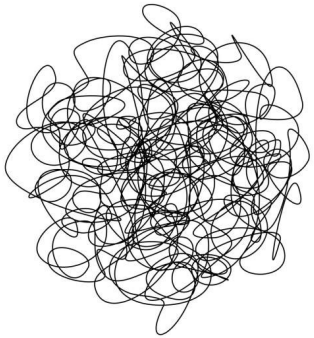
Clinical Notes are:

- Long, lack structure, and fragmented
- Difficult to search across multiple patients

Hard to answer questions like:

- “What treatments worked for similar patients?”
- “Are there patterns across these cases?”

Which leaves clinicians **overwhelmed** and **unable to quickly extract meaningful insights.**



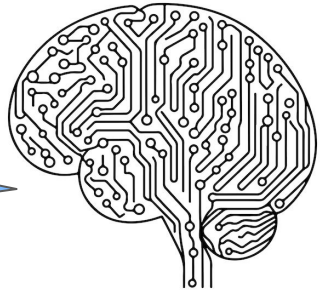
Our Solution: RAG-Powered Clinical Insight Engine

- Retrieval-Augmented Generation
- Combines:
  - Document retrieval (relevant patient notes)
  - LLM Summarization (GPT-4)
- Delivered via simple web interface (Streamlit)

User Query
⬇️
Retrieve relevant notes
⬇️
Generate grounded answer

"What treatments worked for elderly heart failure patients?"

"Diuretics and beta blockers were most common (12 notes)"
Sources: Note IDs 123, 456…



Unlike standard RAG, our system uses intelligent query routing and multi-stage retrieval to improve reliability and context.

How It Works (Pipeline)

Data Processing

1

- Notes chunked by clinical sections (e.g., Assessment, Plan)
- Metadata preserved (patient id, admission id)

Retrieval

2

- Query is first classified before retrieval
- Converts note chunks into embeddings (semantic representations)
- Stores and searches using vector database (FAISS)

Generation Layer

3

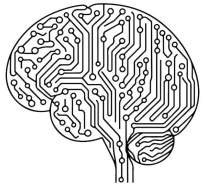
- GPT-4 synthesizes insights from retrieved note segments
- Generates answers using only retrieved evidence and returns source citations

✔ Answers include source citations and confidence scores based on supporting notes

✔ Evaluated using curated clinical QA pairs to verify accuracy

What Makes Our System Smarter Than Standard RAG?

Agentic routing + multi-stage retrieval for reliable answers



Understands the Query First

- LLM query routing
- Classifies Query Type: Patient, Note, Visit, or Semantic
- Uses structured search when possible
- Falls back to semantic search when needed

More reliable than embedding-only retrieval

Retrieves With Context

- Matches relevant sections for precision
- Uses full notes for complete context

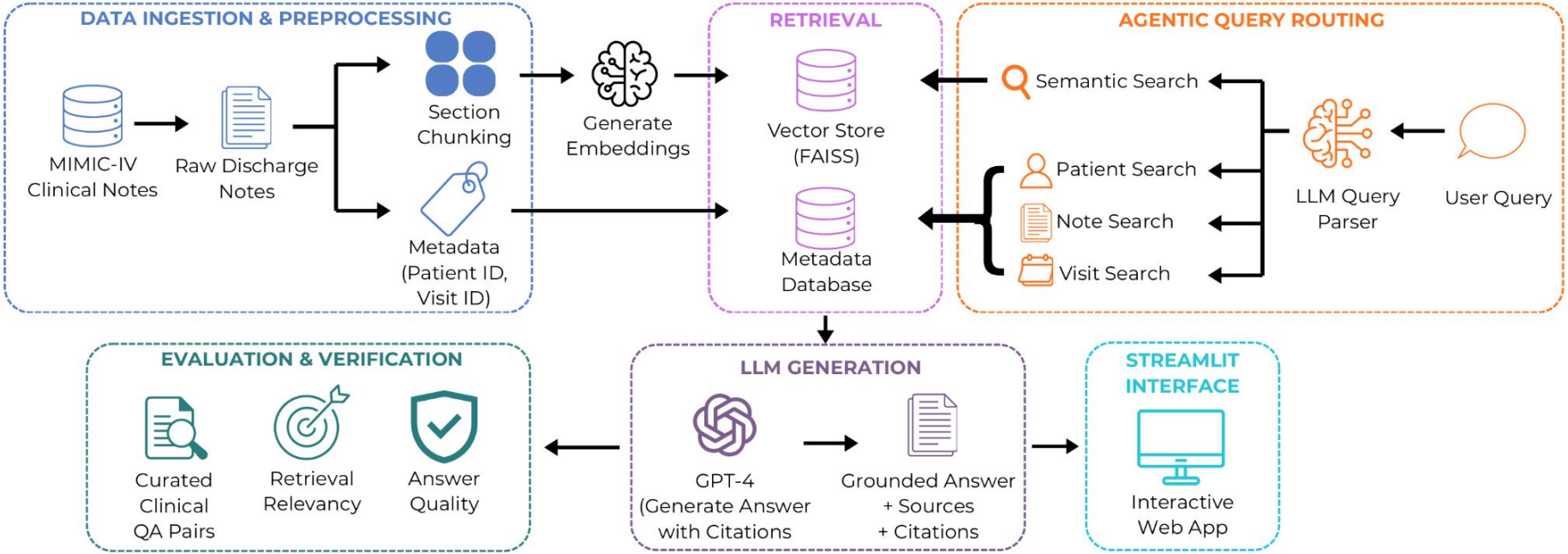
Avoids losing important clinical meaning

Shows Its Work

- Always returns **Source citations**
- Confidence based on supporting notes
- Never answers without evidence

Transparent and trustworthy

Methodology



What This Enables

Population level summarization:

Use Case 1

“What treatments were used for elderly heart failure patients?”

➡ Supports clinical decision making

Patient level summarization:

Use Case 2

“What are this patient’s current active medical problems and how are they being managed?”

➡ Helps identify real-world drug risks across patients

Impact:

- ★ Faster chart review
- ★ Evidence-backed insights
- ★ Scalable across large datasets
- ★ Improved clinical decision making

Live Demo

THANK YOU